NURSADUL MAMUN

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RESEARCH INTEREST

- Audio, Speech and Language Processing
- Speech Enhancement
- Auditory Nerve Modelling
- Cochlear Implants and Hearing aids
- Speech and Speaker Recognition
- Speech Intelligibility and Quality
- Machine Learning and Algorithm Design

EDUCATIONAL QUALIFICATIONS:

PhD in Electrical Engineering — May '18 to Present (Pursuing)
Erik Jonsson School of Engineering and Computer Science
The University of Texas at Dallas, USA.
Thesis Title: Design Speech Enhancement Algorithms for Cochlear Implant Users.

Master of Science in Biomedical Engineering — Sept.'13 to July '15
University Malaya,
Kuala Lumpur, Malaysia.
Result: Distinction
Thesis Title: Prediction of Speech Intelligibility Using a Physiologically Based Model of The Auditory System.

Bachelor of Science in Electrical and Electronic Engineering, — March '08 to Dec. '12
Chittagong University of Engineering and Technology
Chittagong, Bangladesh.
CGPA: 3.89/4 (90.76%.**)**

MAJOR COURSES:

Speech and Speaker Recognition, Machine learning, Pattern Recognition and Machine Learning, Probability, Statistics and Random Variable, Digital Signal Processing, Speech Perception.

RESEARCH EXPERIENCE:

Research Assistant — **May '18- Present**
Cochlear Implant Laboratory ECSN 4.418
Erik Jonsson School of Engineering and Computer Science
The University of Texas at Dallas

Research Activities:

- **Speech Enhancement Algorithm:** Design a Convolutional Neural Network (CNN) based speech enhancement algorithm for cochlear implant users. The algorithm performs speech enhancement in a cochlear filter-bank feature space, a feature-set specifically designed for CI users based on CI auditory stimuli. The algorithm represents a viable option for implementation on the CCi-MOBILE research platform as a pre-processor for CI users in naturalistic environments.
- **Speaker Identification Algorithm:** Design a speaker identification (SID) algorithm to analyzes and quantifies the ability of CI-users to perform speaker identification based on direct electric auditory stimuli. The input speech signal is processed using a CI Advanced Combined Encoder (ACE) signal processing strategy to construct the CI auditory electrodogram and perform SID using electrodogram.

Research Assistant Jan. '13- Feb. '16.
Auditory Neuroscience Laboratory
Department of Biomedical Engineering,

Research Activities:

- **Speech Intelligibility Metric:** Design speech intelligibility metric using a physiological based model of the auditory system. The metric can predict human speech intelligibility for listeners with and without hearing loss both in quiet and noisy conditions. A well-known orthogonal polynomial measure was used as a feature's extractor from the auditory neurogram**.**

- **Robust Gender Classifier:** Design a robust gender classification based on neural responses. This study proposes a gender classification technique using the neural responses of a physiologically-based computational model of the auditory periphery and the performance of the proposed method was evaluated for eight different types of noise.

TEACHING EXPERIENCE:

Assistant Professor 7^{th} Feb. '18 – 7^{th} May '18
Department of Electronics and Telecommunication Engineering
Chittagong University of Engineering and Technology (CUET)
Bangladesh.

Lecturer 24^{th} April '16 – 06^{th} Feb '18
Department of Electronics and Telecommunication Engineering
Chittagong University of Engineering and Technology (CUET)
Bangladesh.

HONOURS AND AWARDS:

- Received all Academic Technical Scholarships for Semester GPA in the Department of Electrical & Electronic Engineering, Chittagong University of Engineering Technology (CUET), Bangladesh. (2008 – 2012)
- Awarded the best poster presenter in International Conference on Innovations in science, Innovation and Technology in Bangladesh for the research on human auditory system. (October, 2016)
- 2019 International Speech Communication Association INTERSPEECH Travel Grant.
- 2019 Conference on Implantable Auditory Prostheses (CIAP) Travel Grant.
- 176^{th} Meeting Acoustical Society of America (ASA) Student Travel Award.

SKILLs

- **Programming Language:** MATLAB, Python, C.
- **Machine Learning:** Classification, Regression, Clustering, LDA, PCA, OLS, WLS, SVD.
- **Machine learning libraries:** Scikit-learn, SciPy, Numpy, Pandas, TensorFlow, Keras.
- **Deep Learning:** DNN, CNN, RNN/LSTM.
- **Circuit Simulation** : P-Spice, DSCH, and MICROWIND.

PROJECT WORKS:

- Design Convolutional Neural Network (CNN)-based Speech Enhancement Algorithms for Cochlear Implant Users.
- Design Speaker Identification Algorithms using Auditory-inspired features for Cochlear Implant Users.
- Self-supervised Speech Enhancement Algorithm using Convolutional Neural Network (CNN).
- Design Wavelet-transform based Signal Denoising for ECG signal.
- Support Vector Machine (SVM)- based arrythmia classification for ECG signal.
- Design Objective Speech Intelligibility Metric for Hearing-impaired Subjects.

PUBLICATIONS:

Journal Papers:

- **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Prediction of Speech Intelligibility Using a Neurogram Orthogonal Polynomial Measure (NOPM)"). IEEE Transaction on Audio, Speech, and Language Processing, 2015 (ISI/SCOPUS *Cited Publication*), vol. 23(4), pp. 760-773.

- **Nursadul Mamun**, S. A. Zilany. "A Spectrogram Orthogonal Polynomial Measure (SOPM) for the Prediction of Speech Intelligibility and Quality" Submitted to JASA, 2020.

- **Nursadul Mamun**, Nahidul Haque Samrat, Choton Kanti Das, Sagor Ghosh." Design and Implementation of a Vehicle Monitoring System for Toll Collection". International Journal of Science & Engineering Research, 2013, V-4, Issue-4.
- Afrin Hossain, Tajrin Jahan Rumky, **Nursadul Mamun**. "Implementation of smart energy meter with two way communication. International Journal of Science & Engineering Research, 2013, V-4, Issue-7.

Conference Papers:

- **Nursadul Mamun,** Soheil Khorram, John H.L. Hansen "Convolutional Neural Network-based Speech Enhancement for Cochlear Implant Recipients, INTERSPEECH 2019, 15-19 September 2019, Graz, Austria**.**
- **Nursadul Mamun,** Ria Ghosh, John H.L. Hansen "Quantifying Cochlear Implant Users' Ability for Speaker Identification using CI Auditory Stimuli, INTERSPEECH 2019, 15-19 September 2019, Graz, Austria**.**
- Khadija Akter**, Nursadul Mamun**, "Predicting Speech Intelligibility with the Regeneration of Envelope from TFS Cues for Hearing Impaired Listeners" EECE**, 7-11 February 2019, Bangladesh.**
- Md. Ibrahim Khalil, **Nursadul Mamun**, Khadija Akter "Robust Text Dependent Speaker Identification Using Neural Responses from the Model of the Auditory System" EECE**, 7-11 February 2019, Bangladesh.**
- John H.L. Hansen, Hussnain Ali, Juliana N. Saba, Ram Charan M. C., **Nursadul Mamun**, Ria Ghosh, Avamarie Brueggeman "CCi-MOBILE: Design and Evaluation of a Cochlear Implant and Hearing Aid Research Platform for Speech Scientists and Engineers" IEEE-EMBS International Conference on Biomedical and Health Information, University of Illinois at Chicago, Chicago, IL, USA, 2019**.**

- Khadija Akter**, Nursadul Mamun** "Prediction of Speech Perception with Recovered Envelope Cues from TFS stimulus For Normal Hearing Listener" ICEEICT 2018, 13-15 September 2018, MIST, Dhaka, Bangladesh**.**
- Taieba Athay, **Nursadul Mamun**, "Predicting Speech Intelligibility Based on Neural Cross-Correlation for Normal Hearing People" 10th International Conference on Electrical and Computer Engineering, 20-22 December 2018, Dhaka, Bangladesh**.**
- Md. Saiful Islam**, Nursadul Mamun** and Muhammad S. Ullah "Speech Based Deception Detection Using Bispectral Analysis", ICGSP, 2018, Australia**.**
- Adnan Basir, **Nursadul Mamun**, "Design of an Effective ECG Filtering Method for Non-linear Noise" International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE 2018), 22-24 November 2018, Dhaka, Bangladesh**.**
- Antora Dev**, Nursadul Mamun** "Design of an EEG-based Brain Controlled Wheelchair for Quadriplegic Patients " IEEE 3rd International Conference for Convergence in Technology (I2CT), 6-8 April 2018, Pune, India**.**
- Shoumya Chowdhury, **Nursadul Mamun**, A. A. Shahjamal Khan, Fahim Ahmed. "Text Dependent and Independent Speaker Recognition Using Neural Responses from the Model of the Auditory System" **11-13 February 2017, Bangladesh.**
- Eftekhar Hossain, **Nursadul Mamun** "Vehicle to Vehicle Communication Using RF and IR Technology" 2nd International Conference on Electrical & Electronic Engineering (ICEEE), 27-29 December 2017, RUET, Rajshahi, Bangladesh**.**
- B. Patwary, Sabrina Abedin, Tasfia Tasbin and **Nursadul Mamun** "Conductivity Based Concentration Measurement Technique For Detecting Adulteration "Proceedings of the International Conference on Mechanical Engineering and Renewable Energy 2017 (ICMERE2017) 18 – 20 December, 2017, Chittagong, Bangladesh**.**
- Taieba Taher, M. A. Haider, **Nursadul Mamun** "Smart Zone Sensing System with Accident Prevention" 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC) 21 - 23 Dec 2017, Dhaka, Bangladesh. (International)
- **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Robust Gender Classification using Neural responses from the Model of the Auditory System" IFESS**, 17-19 September 2014, Malaysia.**
- **Nursadul Mamun**, W. A. Jassim, S. A. Zilany. "Speech-based Gender Classification Using Neurogram Orthogonal Polynomial measure**" RCEE, 04-05 Mar 2014, Malaysia.**
- **Nursadul Mamun**, Nahidul Haque Samrat, Nur Muhammad Khan, Jahirul Islam. "Multi-directional Solar Tracker Using Low Cost Photo Sensor Matrix". International Conference on Informatics, Electronics & Vision (ICIEV), IEEE, 2014.
- Nahidul Hoque Samrat, **Nursadul Mamun**, Shah Mukim Uddin, and Arif Islam. "Case study: Tidal current energy potential in Bangladesh." International Conference on Informatics, Electronics & Vision (ICIEV), pp. 1-5. IEEE, 2013.

Conference Abstracts or Presentations:

- **Nursadul Mamun**, Soheil Khorram, John H.L. Hansen "CCi-MOBILE: Environment-specific speech enhancement with cochlear implant listeners using convolutional neural network, 2019 Conference on Implantable Auditory Prostheses (CIAP), Lake Tahoe, California, USA**.**
- Ria Ghosh, Juliana N. Saba**, Nursadul Mamun**, Hussnain Ali, John H.L. Hansen "CCi-Mobile: moving towards exploring advanced research paradigms for cochlear implant and hearing aid users, 2019 Conference on Implantable Auditory Prostheses (CIAP), Lake Tahoe, California, USA**.**
- Evelyn E. Davies-Venn, **Nursadul Mamun** , Md Hossain , Timothy Kwan, Melanie Putman , MSA Zilany "Comparison of the predictive accuracy of different computational models of auditory perception" 177th Meeting Acoustical Society of America(ASA), 13-17 May 2019, Louisville, Kentucky**, USA.**
- **Nursadul Mamun**, Khadija Akter, Hussnain Ali, John H. L. Hansen "Measuring Speech Perception with Recovered Envelope Cues using the Peripheral Auditory Model" 176th Meeting Acoustical Society of America(ASA), 5-9 November 2018, Victoria, Canada**.**
- Hussnain Ali, **Nursadul Mamun**, John H. L. Hansen, "The CCi-MOBILE VOCODER" 176th ASA, 5-9 November 2018, Victoria, Canada**.**
- S. A. Zilany**, Nursadul Mamun**, W. A. Jassim. "Prediction of Behavioral Speech Intelligibility and Quality using a Computational Model of the Auditory System" 37th Mid-Winter Meeting of the Association for Research in Otolaryngology (ARO), 21-25 Feb 2014, USA.
- Evelyn Davies-Venn**, Nursadul Mamun**, S. A. Zilany. "Computational auditory model of intensity effects on amplified speech perception" 39th Mid-Winter Meeting of the Association for Research in Otolaryngology (ARO), 20-24 Feb 2016, San Diego, California, USA.

Conference Workshops:

- John H.L. Hansen, **Nursadul Mamun**, Ria Ghosh, Juliana N. Saba "Hands-On with CCi-MOBILE: A Cochlear Implant and Hearing-Aid Research Platform, 44th Mid-Winter Meeting (virtual) of the Association for Research in Otolaryngology (ARO), Florida, 21-25 February 2021, USA**.**

- John H.L. Hansen, **Nursadul Mamun**, Ria Ghosh, Juliana N. Saba "CCi-MOBILE: A Cochlear Implant and Hearing-Aid Research Platform, 2019 Conference on Implantable Auditory Prostheses (CIAP), Lake Tahoe, California, USA**.**

ACADEMIC REFEREE

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